



TEST SPEC

Test Spec #:	KCTS0091
Description:	HMI Manual Functional Test
Revision:	CS
Revision Date:	09/27/2024
Revised By:	MMG
Revision History:	Rev A – Initial release Rev B – Adding HMI versions for other control types Rev C – Updated to generic test specification

1.0 Purpose and Scope

- 1.1 This procedure describes the process for performing a Functional Test for the Variable Speed Motor HMI (KCCA0311, KCCA0448, KCCA0231), to test that all buttons, LEDs, LCD segments, LCD backlight LEDs function properly.

2.0 Roles and Responsibilities

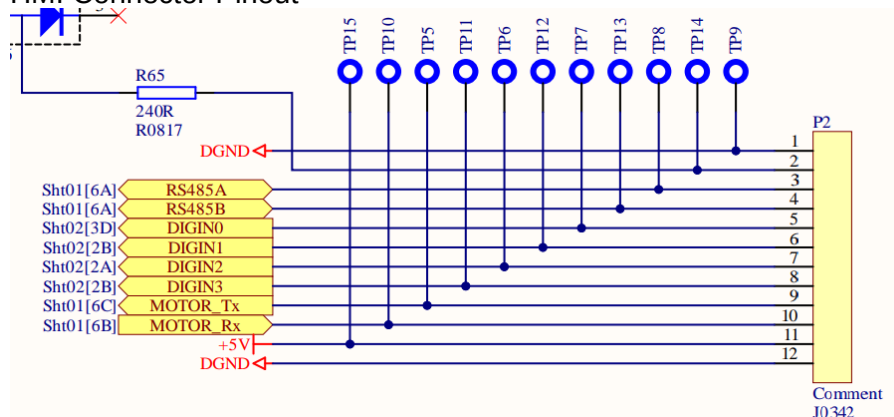
Role	Responsibility
Engineering	Verify test equipment and procedures are being followed.

3.0 Definitions

- 3.1 Motor Control – Variable frequency / variable voltage inverter drive designed to drive an induction motor and human machine interface (HMI) via isolated low voltage
- 3.2 HMI – Human Machine Interface or means for user to input settings
- 3.3 KCCA0278 – Poseidon Motor Control
- 3.4 KCCA0348 – Roadrunner Motor Control
- 3.5 KCCA0330 – Roadrunner 2 Motor Control
- 3.6 KCCA0485 – Coyote Motor Control
- 3.7 KCCA0311 – HMI (NXP)
- 3.8 KCCA0448 – HMI (ST)
- 3.9 KCCA0231 – HMI (Renesas)

4.0 Connections

4.1 HMI Connector Pinout



Pin	Function	Test Point
1	Ground	TP9
2	Output Voltage	TP14
3	RS485A	TP8
4	RS485B	TP13
5	Digital Input 0	TP7
6	Digital Input 1	TP12
7	Digital Input 2	TP6
8	Digital Input 3	TP11
9	RS232 Tx	TP5
10	RS232 Rx	TP10
11	+5V Input	TP15
12	Ground	TP9

4.2 Connection sets

- 4.2.1 +5V Input and Ground to power supply.
- 4.2.2 +5V Input to voltage measurement for software comparison.
- 4.2.3 RS485 to be connected to Test PC
- 4.2.4 RS232 to be connected to either:
 - 4.2.4.1 Corresponding motor controller.
 - 4.2.4.2 Protocol simulation on Test PC.
- 4.2.5 Digital Inputs to relay board allowing controlled connections to Output Voltage
- 4.2.6 Output Voltage to voltage measurement. Range = 0-24V.

5.0 Tests

5.1 Serial Number Scan

- 5.1.1 UUT serial number to be scanned and recorded. Traceability is required to the component level from the serial number.

5.2 Voltage Measurement

- 5.2.1 Power is provided to the UUT
- 5.2.2 Voltage is verified on Voltage Output

5.3 Communication Check

- 5.3.1 Use a firmware version request via RS485 connection to confirm communications are functioning.

5.4 Verify LED Functionality

- 5.4.1 Command all LEDs to be on.
- 5.4.2 Verify intensity level on all LEDs.

5.5 Verify LCD Functionality

- 5.5.1 Command text to the LCD display.
- 5.5.2 Verify all segments are functional and backlight intensity level.

5.6 Verify Button Functionality

5.6.1 Actuate each button individually.

5.6.2 Verify button state via RS485.

5.7 Verify digital inputs

5.7.1 Short each digital input pin to Output voltage individually.

5.7.2 Verify digital input state via RS485.